Deploying and Monitoring a SIEM by using Wazuh to Detect Intrusion Attempts

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Introduction

Wazuh is an open-source SIEM (Security Information and Event Management) platform used for real-time monitoring, log analysis, and threat detection. It integrates various security features including file integrity monitoring, vulnerability detection, intrusion detection, and behavioral analysis.

The objective of this task is to implement a complete SIEM system using Wazuh by installing its core components (Wazuh Manager, Elasticsearch, Kibana) and ensuring the system is capable of detecting suspicious activities. The task includes executing a simulated attack to evaluate Wazuh's effectiveness in identifying and alerting on threats. Kali Linux was used as the virtual environment to provide a secure and controlled testing space.

Environment

-Operating System: Kali Linux (Virtual Machine)

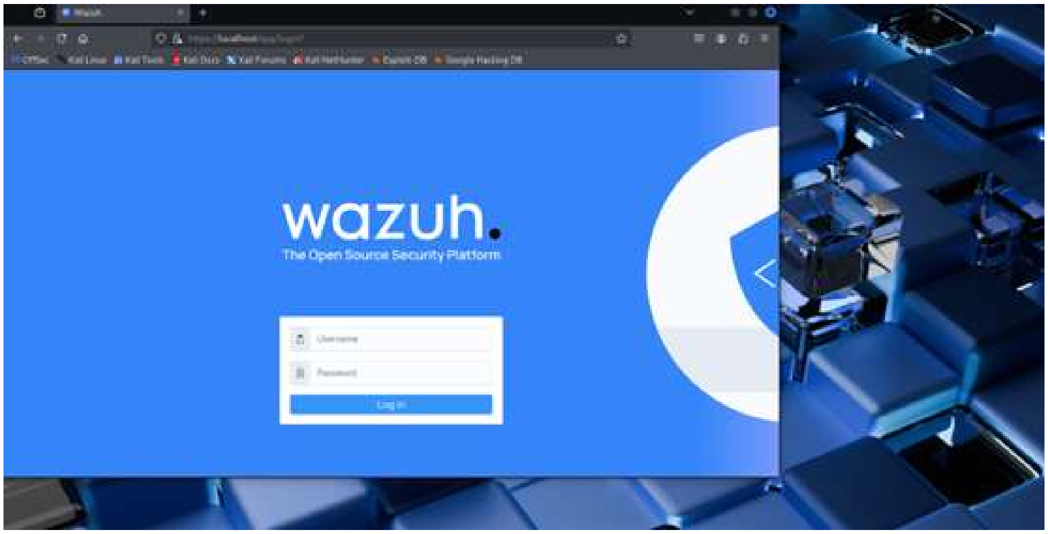
-Analysis Tools: Wazuh Dashboard, Kibana

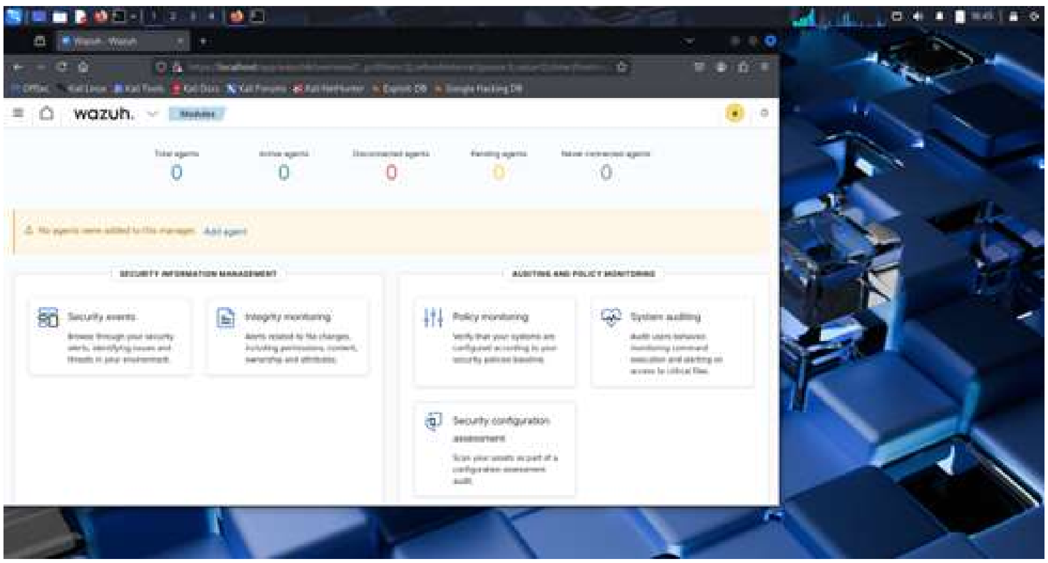
-Network: Localhost

-Wazuh SIEM Installation Script (with Elasticsearch & Kibana)

```
#!/bin/bash

# 1. Update the system

sudo apt update && sudo apt upgrade -y


# 2. Install Wazuh Manager

curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o
/usr/share/keyrings/wazuh-archive-keyring.gpg

echo "deb [signed-by=/usr/share/keyrings/wazuh-archive-keyring.gpg]
https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee
/etc/apt/sources.list.d/wazuh.list

sudo apt update

sudo apt install wazuh-manager -y

sudo systemctl enable --now wazuh-manager
```

```
# 3. Install Elasticsearch 7.17.0

curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --dearmor -o
/usr/share/keyrings/elastic-keyring.gpg

echo "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg]
https://artifacts.elastic.co/packages/7.x/apt stable main" | sudo tee
/etc/apt/sources.list.d/elastic-7.x.list

sudo apt update

sudo apt install elasticsearch=7.17.0 -y

sudo systemctl enable --now elasticsearch

sleep 20  # Give Elasticsearch time to initialize


# 4. Install Kibana 7.17.0

sudo apt install kibana=7.17.0 -y

sudo systemctl enable --now kibana


# 5. Output access information

echo ""

echo " Installation completed!"

echo "Access the web interfaces:"

echo " 🔹 Wazuh Dashboard: https://<your-ip>:5601"

echo " 🔹 Kibana: http://localhost:5601"

echo ""

echo " Make sure to configure Kibana and Wazuh integration for alert visualization."
```



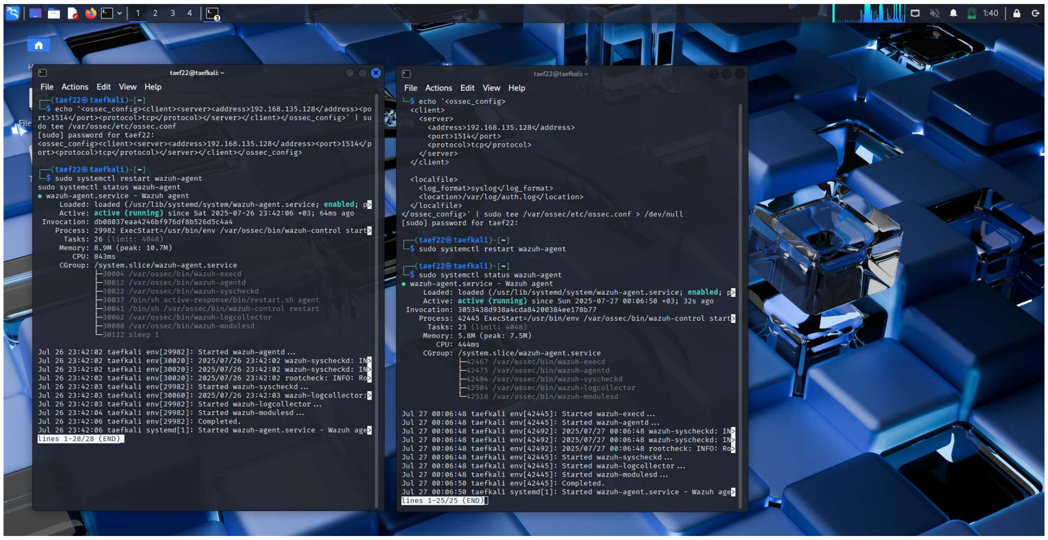


Agent Installation & Integration

The Wazuh agent was installed using the following command:

sudo apt install wazuh-agent -y

The agent was successfully configured and registered with the Wazuh Manager. It appeared in the Wazuh Dashboard and started sending system logs.



Alert Verification

File Integrity Monitoring

A file was intentionally modified to trigger an integrity check alert. The alert appeared in the Wazuh Dashboard as expected.

Vulnerability Detection

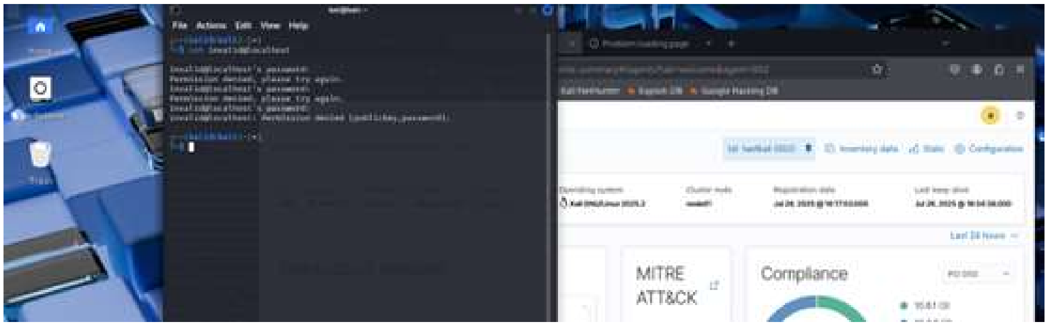
Wazuh scanned installed packages and reported vulnerabilities with Medium and Low severity. These alerts confirmed that the vulnerability detection module was functioning.

Simulated Attack

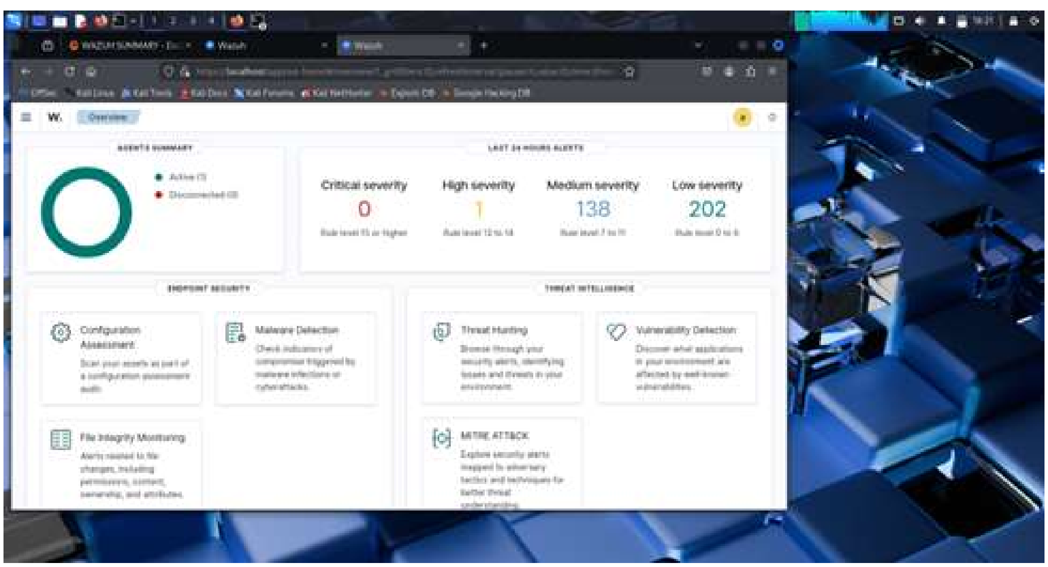
A brute-force SSH login attempt was simulated using:

for i in {1..7}; do ssh testuser@localhost; done

Wazuh successfully detected multiple authentication failures and generated alerts in the Dashboard.



The attack was successfully detected



The Wazuh Dashboard displays real-time agent status and recent alerts categorized by severity. This screenshot confirms that one agent is actively connected and that the system has detected

multiple medium and low severity alerts over the past 24 hours, demonstrating successful monitoring and detection capabilities.

Final Result

The mission was successfully completed:

- All components were installed and configured properly.
- Wazuh Dashboard showed real-time monitoring.
- Alerts for integrity violations, vulnerabilities, and attacks were successfully triggered and visualized.

Conclusion

The Wazuh SIEM system proved to be effective in monitoring, detecting, and visualizing system-level threats using a realistic simulated scenario within a local Kali Linux VM environment.